

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

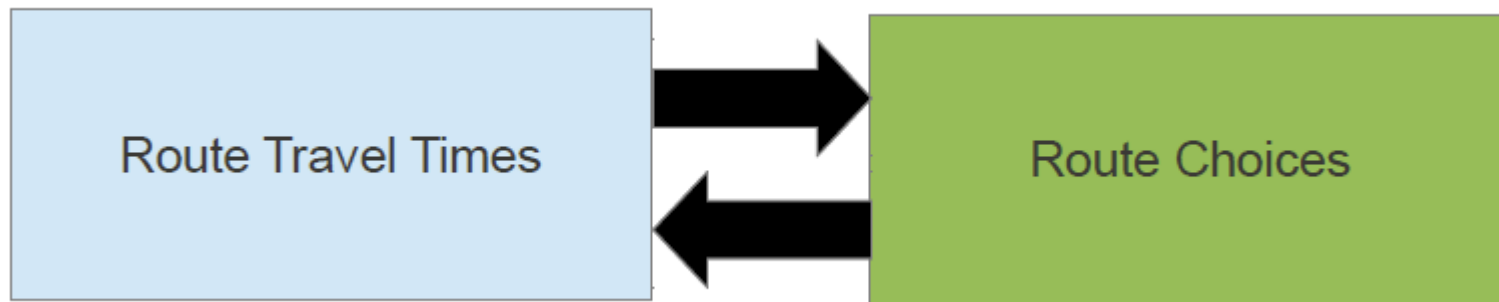
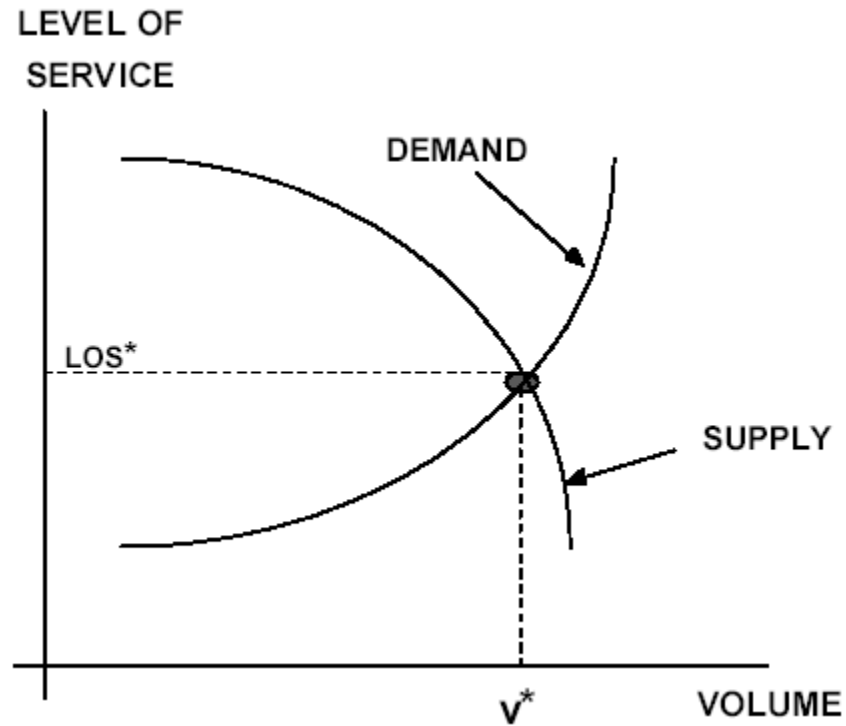
Traffic Assignment

Lecture Week 12

Dr. Mehmet Yildirimoglu
School of Civil Engineering
The University of Queensland
m.yildirimoglu@uq.edu.au

- Introduction to traffic assignment / equilibrium
- Path choice models
 - Deterministic vs. Probabilistic
- Static traffic assignment
 - User Equilibrium (UE)
 - System Optimum (SO)
 - Methods to reach equilibrium
- Dynamic traffic assignment
 - Dynamic User Equilibrium (DUE),
 - Dynamic System Optimum (DSO)
- Departure time choice

Introduction

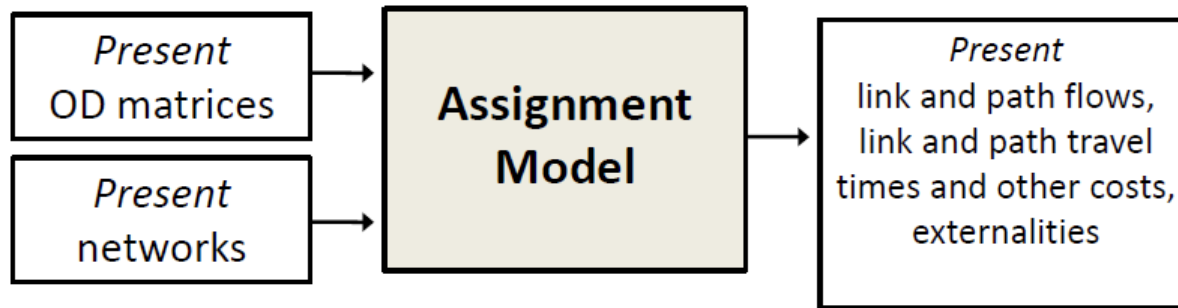




Static traffic assignment

Applications of Assignment models

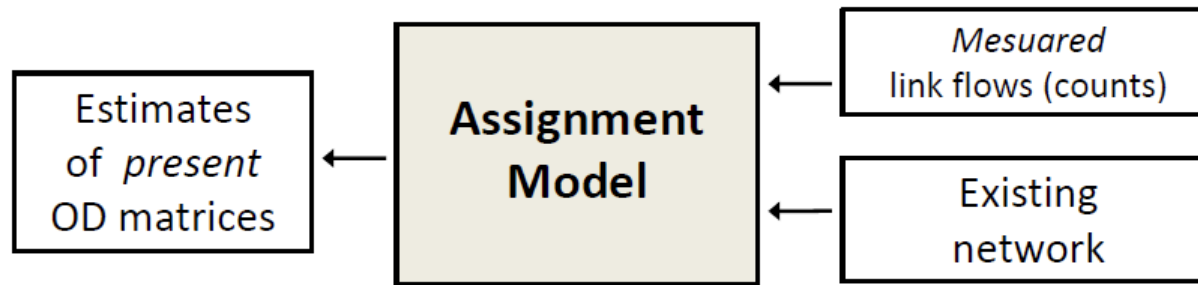
- As estimators of the current state of the transport systems



- A good match between observed and simulated traffic conditions is needed to guarantee a well calibrated simulation model
- The results from the assignment model can complement other traffic observations (e.g., link flows) as such measures are not available for all network components

Applications of Assignment models

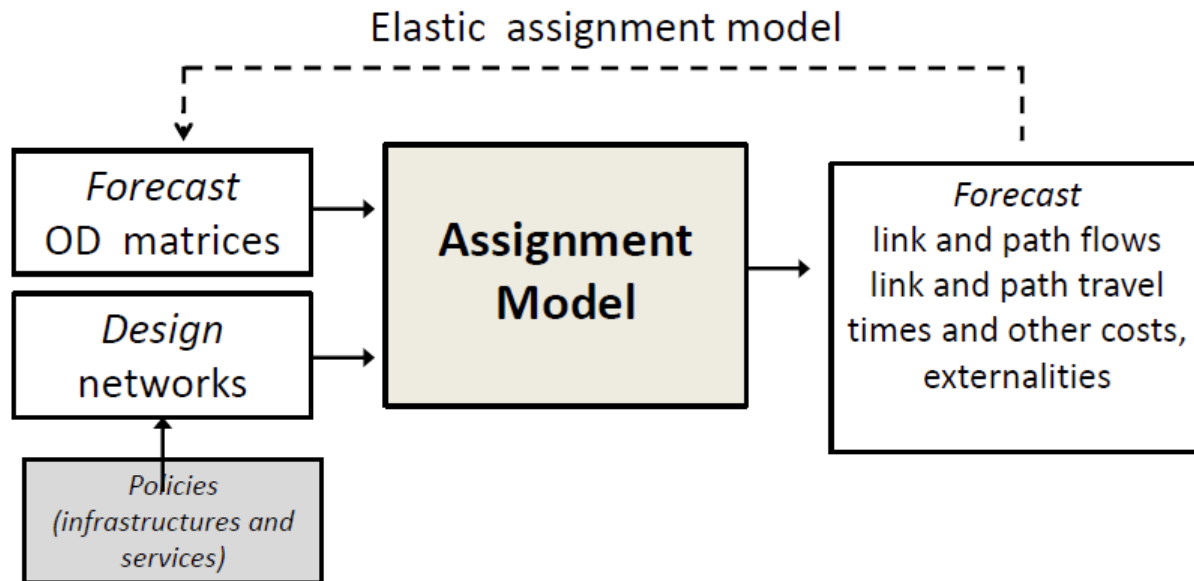
- For the estimation of current travel demand



- If prior OD matrices are not reliable, assignment models can be useful to estimate present OD matrices
- Prior ODs usually depend on historical surveys and may not reflect current traffic state

Applications of Assignment models

- For evaluating the changes in the transport systems



- Traffic assignment considers a trade-off between the demand (the number of vehicles willing to drive along a network link) and supply (the available capacity of that network link)

- This trade-off implies that there are three building blocks
 - A network of nodes and links
 - A link performance function (speed-flow or time-flow) that expresses the trade-off between traffic volume and travel time. This function may be static or dynamic depending on the application.
 - A route choice model that controls the assignment of the demand to the network links

- Given the demand, the network and the link performance functions
 - **All or Nothing** assignment with no road capacity constraints
 - **User Equilibrium (UE)** with capacity constraints
 - **Stochastic User Equilibrium (SUE)** with capacity and stochastic effects

- Dynamic extensions
 - **Dynamic** User Equilibrium
 - **Dynamic** Stochastic User Equilibrium

All or Nothing assignment

- Assumptions
 - Traffic flows freely without delays (no congestion)
 - Travellers have full knowledge of the network
 - All travellers are equal and have the same preferences
- The principle is then that all travellers choose the shortest path from their origin to their destination
- The advantage is that the assignment model is really fast, given that it is a search for shortest paths from multiple origins to multiple destinations
- The disadvantage is that the assignment model does not consider congestion, it is unrealistic and it is unusable for evaluating planning projects

- Given the demand, an equilibrium principle exists so that the system can be described by an equilibrium state
- An equilibrium model needs to be given to represent the state where demand and supply reach the equilibrium itself
- The All or Nothing assignment model is not an equilibrium
 - The model allows for infinite demand to be on one path
 - The model converges in 1 iteration (shortest path search)

Wardrop's equilibrium principles

In 1952, Wardrop stated two principles that formalize different notions of equilibrium

- 1. Principle

“Under equilibrium conditions traffic arranges itself in congested networks in such a way that no individual traveller can improve their path cost by switching routes”

- 2. Principle

“Under equilibrium conditions traffic should be arranged in congested networks in such a way that the total travel cost is minimised”

Wardrop's equilibrium principles

- Wardrop's first principle is a UE (user equilibrium)
- **Descriptive:** it describes how individuals travel through an urban transportation system given the components of that system
- Each traveller minimises his/her own travel cost, in line with the utility maximisation principle (i.e., realistically travellers are selfish)
- The UE implies mathematically that, if T_{ijr}^* is the traffic flow at equilibrium for OD pair $\{i,j\}$ on a given route r , and c_{ij}^* is the travel cost at equilibrium for the same OD pair, then

$$T_{ijr}^* = 0 \ \& \ c_{ijr} > c_{ij}^*$$

$$T_{ijr}^* > 0 \ \& \ c_{ijr} = c_{ij}^*$$

- The UE implies that the travel times on all used routes are equal and minimal

Wardrop's equilibrium principles

- Wardrop's second principle is a SO (system optimum)
- **Normative:** attempt to determine the optimal system configuration
- Force route choices such that the total travel time on the network is minimised
- From the planners' perspective, this is the most beneficial principle although selfish travellers would change route for a better one
- As UE and SO are not always the same, planners will push the UE as close as possible to the SO with a set of instruments
 - Road pricing ([video](#))
 - Renovated or new infrastructure
 - Public transport subsidising

- By definition *system optimal* flows are not stable. Travelers will always be tempted to switch to non-system optimal routes.
- Therefore system optimal flows are **not realistic of actual traffic**.
- However, system optimal flows do provide useful comparisons with the more realistic user equilibrium traffic forecasts.

- SUE means that travellers cannot improve their **perceived** travel cost by unilaterally changing their routes
- All routes are used and the travel times are not equal
- Probabilistic path choice models

Wardrop's equilibrium principles

- UE formulated as a mathematical program by Beckmann, 1956

$$\text{Min } Z = \sum_a \int_{0, x_a} t_a(x) dx$$

st.

$$\sum_k f_{k,i,j} = T_{i,j}$$

$$x_a = \sum_k f_{k,i,j} \delta_{a,k,i,j}$$

$$f_{k,i,j} \geq 0 \quad \forall k,i,j$$

$$x_a \geq 0 \quad \forall_a$$

- Recall that UE means that the travel times of used routes are equal

Wardrop's equilibrium principles

- SO was formulated as a mathematical program by Sheffi

$$\text{Min } Z = \sum_a x_a t_a(x_a)$$

st.

$$\sum_k f_{k,i,j} = T_{i,j}$$

$$x_a = \sum_k f_{k,i,j} \delta_{a,k,i,j}$$

$$f_{k,i,j} \geq 0 \quad \forall k,i,j$$

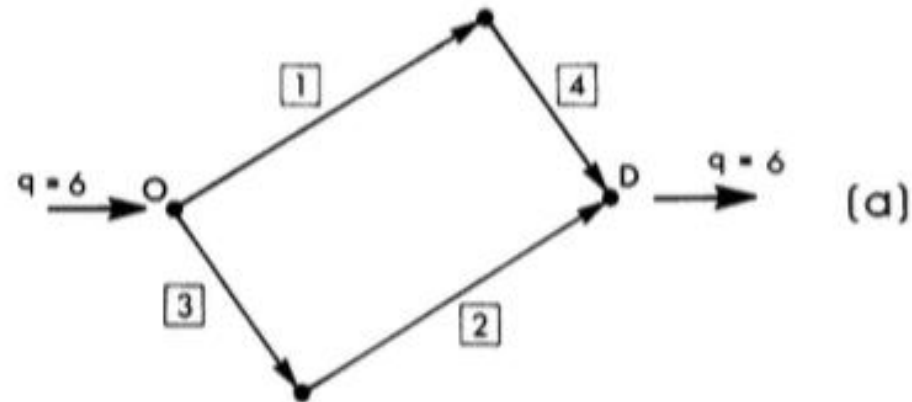
$$x_a \geq 0 \quad \forall_a$$

- Recall that SO means that the total travel time is minimum, and the total travel time is the sum of the travel time on each link multiplied by the traffic flow on that link

Exercise

- Two routes connect a city and a suburb. During the peak-hour morning commute, a total of 4500 vehicles travel from the suburb to the city. Route 1 has a 60 km/h speed limit and 6km in length. route 2 is 3 km in length with 45km/h speed limit. Studies show that the travel time on route 1 increases four minutes for every additional 1000 vehicles added. Minutes of travel time on route 2 increase with the square of the number of vehicles expressed in thousands of vehicles per hour. Determine user equilibrium travel times.

Example of UE and SO calculation



Performance Data

$$\begin{aligned}t_1(x_1) &= 50 + x_1 \\t_2(x_2) &= 50 + x_2 \\t_3(x_3) &= 10x_3 \\t_4(x_4) &= 10x_4\end{aligned}$$

Path Definition



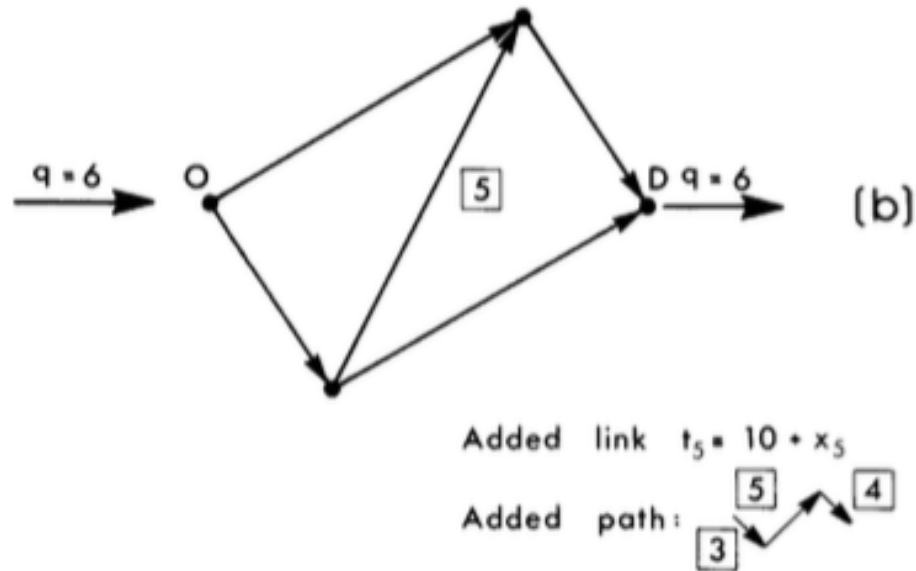
Example of UE calculation

- The answers to the problems are
 - The four travel times t_1, t_2, t_3, t_4
 - The four link volumes x_1, x_2, x_3, x_4
- User equilibrium means that the travel time on the two routes is equal
 - In the specific case $t_1 + t_4 = t_2 + t_3$
 - Write the travel times as a function of the four volumes x_1, x_2, x_3, x_4 given the link performance functions
 - Write the conservation of the traffic volumes at the nodes (in the specific case $x_1 = x_4$ and $x_2 = x_3$)
 - Write the total demand (in the specific case $x_1 + x_2 = 6$)
 - This is a system of four equations in four unknowns that can be easily solved

Example of SO calculation

- The answers to the problems are
 - The four travel times t_1, t_2, t_3, t_4
 - The four link volumes x_1, x_2, x_3, x_4
- System optimum means that the total travel time on the network needs to be written as a function of time and volumes and then be minimised
 - In the specific case $t_1x_1 + t_2x_2 + t_3x_3 + t_4x_4$
 - Write the travel times as a function of the four volumes x_1, x_2, x_3, x_4 given the link performance functions
 - Write the conservation of the traffic volumes at the nodes (in the specific case $x_1 = x_4$ and $x_2 = x_3$)
 - Write the total demand (in the specific case $x_1 + x_2 = 6$)
 - This is a function with three equations and four unknowns
 - The minimum is obtained by deriving the resulting function by the only unknown left and having the derivative equal to zero

Example of UE and SO calculation



Example of UE and SO calculation

- In the second case the total travel time in the network under UE conditions was worse even though the demand was the same and the network improved: this is the **Braess paradox** and it is related to the travellers being selfish
- From a more general perspective, **Braess paradox** emphasizes the importance of a careful and systematic analysis of investments in urban networks. Not every addition of capacity can bring about all the anticipated benefits and in some cases, the situation may be worsened.

Method of successive averages (MSA)

- **Step 0: Initialization.** Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on $t_a = t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter $n = 1$
- **Step 1: Update.** $t_a^n = t_a(x_a^n)$
- **Step 2: Direction finding.** Perform all-or-nothing assignment based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3: Define an updating parameter.** Usually $\phi = 1/n$.
- **Step 4: Move.** $x_a^{n+1} = x_a^n + \phi(y_a^n - x_a^n)$, $\forall a$
- **Step 5: Convergence check.** If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

$$\|x_a^{n+1} - x_a^n\| \leq \varepsilon$$

Frank-Wolfe Assignment

- **Step 0: Initialization.** Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on $t_a = t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter $n = 1$
- **Step 1: Update.** $t_a^n = t_a(x_a^n)$
- **Step 2: Direction finding.** Perform all-or-nothing assignment based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3: Line search.** Find α_n that solves:

$$\min_{0 \leq \alpha \leq 1} \sum_a \int_0^{x_a^n + \alpha(y_a^n - x_a^n)} t_a(w) dw$$

- **Step 4: Move.** $x_a^{n+1} = x_a^n + \alpha_n(y_a^n - x_a^n)$, $\forall a$
- **Step 5: Convergence check.** If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

- **Step 0: Initialization.** Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on $t_a = t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter $n = 1$
- **Step 1: Update.** $t_a^n = t_a(x_a^n)$
- **Step 2: Direction finding.** Perform a **stochastic assignment** based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3: Define an updating parameter.** Usually $\phi = 1/n$.
- **Step 4: Move.** $x_a^{n+1} = x_a^n + \phi(y_a^n - x_a^n)$, $\forall a$
- **Step 5: Convergence check.** If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

$$\|x_a^{n+1} - x_a^n\| \leq \varepsilon$$

Next

This week

- Tutorial
- Problem set released